

Lecture 10

Let us try to understand the well-definedness of the operation defined on G/H by an example.

G group. H subgroup of G .

G/H is the set of all left cosets of H in G .

A natural way to define a binary operation on G/H is by saying

$$(g_1 H)(g_2 H) = g_1 g_2 H.$$

Now let us consider the following example.

$$G = S_3 = \{e, (12), (13), (23), (123), (132)\}$$

Consider the subgroup $H = \{e, (12)\}$.

Let's compute all the left cosets of H in S_3 .

$$(13)H = \{(13)e, (13)(12)\} = \{(13), (123)\}$$

$$(123)H = \{(123)e, (123)(12)\} = \{(123), (13)\}$$

$$(23)H = \{(23), (132)\}$$

$$(132)H = \{(132), (23)\}.$$

$$\text{So } G/H = \{\cancel{eH} \mid gH \mid g \in S_3\}$$

$$= \{\{e, (12)\}, \{(13), (123)\}, \{(23), (132)\}\}.$$

Now let us apply the definition on G/H .

$$(13)H \cdot (23)H = (13)(23)H = (132)H.$$

$$(13)H = (123)H$$

$$(23)H = (132)H$$

$$(123)H \cdot (132)H = (123)(132)H = eH = H.$$

Even though $(13)H = (123)H$
& $(23)H = (132)H$.

$$(132)H \neq H.$$

So the operation is not well-defined!!!

Theorem: Let G be a group. $H \trianglelefteq G$. Then the operation on G/H defined by

$$(g_1H)(g_2H) = g_1g_2H \text{ is well defined.}$$

Proof: Let $g_1H = g'_1H$ & $g_2H = g'_2H$

Need to prove $(g_1H)(g_2H) = (g'_1H)(g'_2H)$

i.e. $g_1g_2H = g'_1g'_2H$.

~~DD~~ Enough to prove $(g_1g_2)^{-1}g'_1g'_2 \in H$.

$$\text{As } g_1H = g'_1H \Rightarrow g_1^{-1}g'_1 \in H$$

$$g_2H = g'_2H \Rightarrow g_2^{-1}g'_2 \in H$$

$$\text{So } (g_1 g_2)^{-1} g'_1 g'_2$$

$$= g_2^{-1} g_1^{-1} g'_1 g'_2$$

$$= \underbrace{g_2^{-1} g_1^{-1} g'_1 g_2}_{\in H} \underbrace{g_2^{-1} g'_2}_{\in H}$$

$$g_1^{-1} g'_1 \in H \quad \& \quad H \trianglelefteq G \Rightarrow g_2^{-1} (g_1^{-1} g'_1) g_2 \in H.$$

$$g_2^{-1} g'_2 \in H \quad \text{Hence } (g_1 g_2)^{-1} g'_1 g'_2 \in H$$

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$$\Rightarrow g_1 g_2 H = g'_1 g'_2 H.$$

So the binary operation is well defined.

Theorem: Let G be a group & $H \trianglelefteq G$. Then

$(G/H, \cdot)$ is a group.

Proof: We have proved that the above operation is well defined.

Let $g_1, g_2, g_3 \in G$.

$$((g_1 H)(g_2 H)) g_3 H = (g_1 g_2) H g_3 H$$

$$= (g_1 g_2) g_3 H$$

$$= g_1 (g_2 g_3) H \quad [\because G \text{ associative}]$$

$$= (g_1 H)(g_2 g_3 H)$$

$$= (g_1 H)(g_2 H g_3 H)$$

$\therefore$ on G/H is associative.

$eH = H$ is the identity element in G/H as

$$(gH) \cdot (eH) = gH = (eH) (gH).$$

for each $gH \in G/H$ we have $g^{-1}H \in G/H$

$$\hookrightarrow (gH)(g^{-1}H) = eH = (g^{-1}H)(gH)$$

So each element of G/H has an inverse.

$\therefore (G/H, \cdot)$ is a group.

Remark: If G is an additive group & $H \leq G$, then

$$G/H = \{g+H \mid g \in G\}.$$

We define an operation same as before by

$$(g_1+H) + (g_2+H) = (g_1+g_2)+H.$$

One can easily show $(G/H, +)$ is also a group.

This is called the quotient group or factor group.

Example: $G = (\mathbb{Z}, +)$, let $H = 2\mathbb{Z}$, $H \leq G$.

Since G is abelian, H is also normal in $\mathbb{Z}$.

$$G/H = \{H, 1+2\mathbb{Z}\}.$$

Then G/H is a group under the above operation.

Result: Let G be an abelian group. $H \trianglelefteq G$. Then G/H is also abelian.

Proof: Let g_1H & g_2H are two elements of G/H .

$$\begin{aligned}\text{Then } (g_1H) \cdot (g_2H) &= g_1g_2H \\ &= g_2g_1H \quad [\because G \text{ is abelian}] \\ &= (g_2H)(g_1H).\end{aligned}$$

Hence G/H is abelian.

Result: Let G be a finite group & $H \trianglelefteq G$. Then

$$|G/H| = \frac{|G|}{|H|}.$$

Proof: G finite, $H \trianglelefteq G \Rightarrow G/H$ is a group.

$$|G/H| = \# \text{ distinct cosets of } H \text{ in } G = [G:H] = \frac{|G|}{|H|}.$$

Group homomorphism: Let $(G, \star)$ & $(G', \star')$ be two groups.

A mapping $f: G \rightarrow G'$ is called a group homomorphism if

$$f(g_1 \star g_2) = f(g_1) \star' f(g_2) \quad \forall g_1, g_2 \in G.$$

① If f is injective, then it is called monomorphism.

② If f is surjective, then it is called epimorphism.

③ If f is bijective, then it is called isomorphism.

Example: ① ~~Def~~ let $G = G' = (\mathbb{Z}, +)$.

Define $f: \mathbb{Z} \rightarrow \mathbb{Z}$.

$$f(n) = 2n.$$

Then
$$f(m+n) = 2(m+n) = 2m + 2n = f(m) + f(n)$$
$$\forall m, n \in \mathbb{Z}.$$

$\therefore f$ is a group homomorphism.

Since f is injective, it is monomorphism.

② let $G = (GL(n, \mathbb{R}), \cdot)$ $G' = (\mathbb{R}^*, \cdot)$

Define: $f: GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$
 $f(A) = \det A.$

$$f(AB) = \det(AB) = (\det A)(\det B) = f(A) \cdot f(B) \quad \left(\forall A, B \in GL(n, \mathbb{R}) \right)$$

Hence f is a homomorphism.

f is surjective as for each $a \in \mathbb{R}^*$,

$$A = \begin{bmatrix} a & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \text{ has } \det A = a.$$

So f is an epimorphism.

Example 3. $G = (\mathbb{Z}, +)$, $G' = (\mathbb{R}^*, \cdot)$

Define $f: \mathbb{Z} \rightarrow \mathbb{R}^*$

$$f(n) = 2^n.$$

$$f(m+n) = 2^{m+n} = 2^m \cdot 2^n = f(m) \cdot f(n), \quad \forall m, n \in \mathbb{Z}.$$

Hence f is a group homomorphism.

4. $G = \mathbb{Z}_7 = G'$ with addition.

Define $f: \mathbb{Z}_7 \rightarrow \mathbb{Z}_7$

$$f([a]) = [2a]$$

$$\begin{aligned} f([a] + [b]) &= f([a+b]) = [2(a+b)] \\ &= [2a] + [2b] = f([a]) + f([b]) \\ &\quad \forall [a], [b] \in \mathbb{Z}_7 \end{aligned}$$

Hence f is a group homomorphism.

f is also both injective & surjective. (Check!!!)

So f is bijective.

So f is an isomorphism.

Two groups G & G' are called isomorphic if there exists an isomorphism between them i.e.

$f: G \rightarrow G'$ an isomorphism.

And we write $G \cong G'$.

Proposition: Let $(G, \cdot)$ & $(G', \cdot)$ be two groups & $f: G \rightarrow G'$ be a group homomorphism. Then the following statements are true.

- ① $f(1) = 1'$ where 1 & $1'$ are identity of G & G' respectively.
- ② $f(g)^{-1} = f(g^{-1}) \quad \forall g \in G.$

Proof: (Note: from now on we use the $\cdot$ to denote the binary operation on both G & G' . Keep in mind that they are different).

① $1 \cdot 1 = 1$ in G .

So $f(1 \cdot 1) = f(1)$

$\Rightarrow f(1) \cdot f(1) = f(1)$ [$\because f$ is a homomorphism]

$\Rightarrow f(1)^{-1} (f(1) \cdot f(1)) = f(1)^{-1} f(1)$

$\Rightarrow f(1) = 1'$ [$\because f(1)^{-1} f(1) = 1'$ in G']

② $f(g) \cdot f(g^{-1}) = f(g \cdot g^{-1})$ [$\because f$ homomorphism]
 $= f(1) = 1'$

Similarly $f(g^{-1}) \cdot f(g) = 1'$

$\Rightarrow f(g)^{-1} = f(g^{-1}).$

Let $f: G \rightarrow G'$ be a group homomorphism.

Define $\text{Ker } f = \{g \in G \mid f(g) = 1'\}$ (This is called the kernel of f)

$$\text{Im } f = \{f(g) \mid g \in G\}.$$

Theorem: $\text{Ker } f$ is a subgroup of G & $\text{Im } f$ is a subgroup of G' . Moreover $\text{Ker } f \trianglelefteq G$.

Proof: Let $a, b \in \text{Ker } f$.

Need to prove $ab^{-1} \in \text{Ker } f$

$$a \in \text{Ker } f \Rightarrow f(a) = 1'$$

$$b \in \text{Ker } f \Rightarrow f(b) = 1'$$

$$f(ab^{-1}) = f(a) \cdot f(b^{-1}) \quad [\because f \text{ homomorphism}]$$

$$= f(a) f(b)^{-1} \quad \text{by } \textcircled{2}$$

$$= 1' \cdot (1')^{-1}$$

$$= 1'$$

$$\Rightarrow ab^{-1} \in \text{Ker } f. \Rightarrow \text{Ker } f \text{ is a subgroup of } G.$$

Let $g \in G$ & $a \in \text{Ker } f$.

To prove $\text{Ker } f \trianglelefteq G$, it is enough to prove

$$gag^{-1} \in \text{Ker } f.$$

$$f(gag^{-1}) = f(g) f(a) f(g^{-1}) \quad [\because f \text{ homomorphism}]$$

$$= f(g) \cdot 1' \cdot f(g)^{-1} \quad [\because a \in \text{Ker } f \text{ & } \textcircled{2}]$$

$$= 1'$$

$$\Rightarrow gag^{-1} \in \text{Ker } f \Rightarrow \text{Ker } f \trianglelefteq G.$$

$$\text{Im } f \subseteq G'.$$

To prove $\text{Im } f \leq G'$, it is enough to prove,

$$\forall a, b \in \text{Im } f, \quad ab^{-1} \in \text{Im } f.$$

$$a \in \text{Im } f \Rightarrow a = f(g) \text{ for some } g \in G.$$

$$b \in \text{Im } f \Rightarrow b = f(g') \text{ for some } g' \in G.$$

$$\text{So } ab^{-1} = f(g) f(g')^{-1}$$

$$= f(g) f(g'^{-1}) \quad [\text{by } \textcircled{2}]$$

$$= f(gg'^{-1}) \quad [\because f \text{ homomorphism}]$$

$$\Rightarrow ab^{-1} \in \text{Im } f.$$

$$\text{Hence } \text{Im } f \leq G'.$$